

AI-Assisted Clinical Data Interface

From Research Questions to Research-Ready Data

The challenge is not only accessing complex clinical data - it is knowing what the data can scientifically support.

THE PROBLEM:

Large clinical research datasets such as the Childhood Cancer Survivor Study (CCSS) and the St. Jude Lifetime Cohort (SJLIFE) contain rich longitudinal information, but investigators often need substantial dataset-specific knowledge to determine:

Why is this hard?

- What variables exist?
- Where are they stored?
- Can the dataset answer the scientific question?
- How should structured and narrative data be combined?

Who feels the friction?

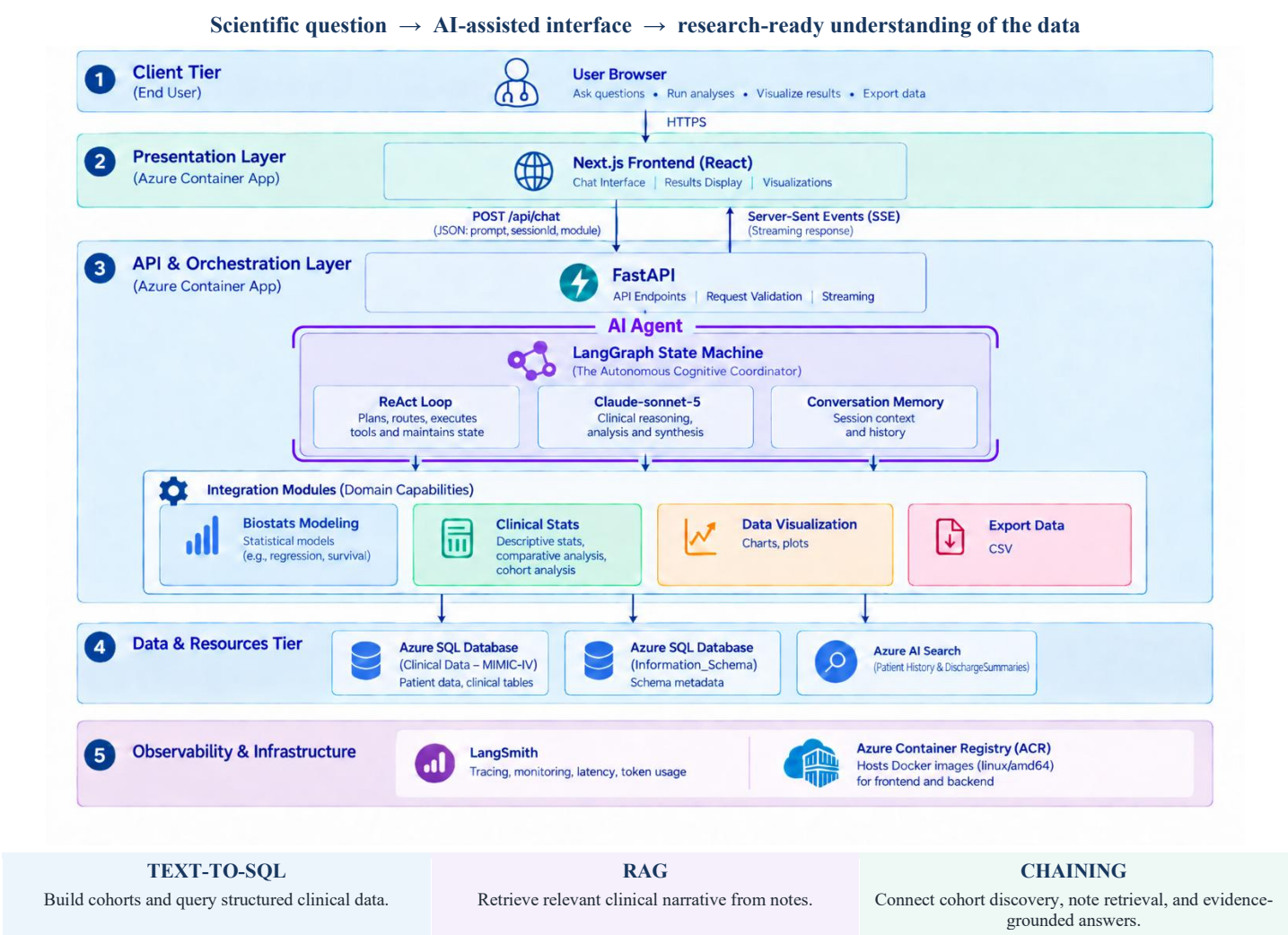
Investigators & clinicians
Need to know what questions the data can support.

Biostatisticians & analysts
Need to know where the information needed lives and whether enough usable data is available.

OBJECTIVE

Develop an AI-assisted interface that helps investigators and clinicians translate scientific or clinical questions into relevant data needs, while enabling biostatisticians and data analysts to identify where those data are located, what information is available, and whether the dataset can adequately support the proposed research question.

OUR APPROACH



What This Looks Like in Practice

One scientific question → the right cohort, variables, sources, and evidence

EXAMPLE USE CASE

“Are early radiographic findings in hospitalized pneumonia patients associated with ICU transfer or in-hospital mortality?”	
Cohort	Hospitalized adults with pneumonia
Structured data	Demographics, diagnoses, laboratory data, microbiology, ICU transfer, mortality
Narrative data	Discharge summaries and radiology reports, when available
Research-ready output	Relevant variables and tables, cohort logic, supporting notes, and an evidence-grounded view of whether the dataset can support the question

WHY THIS IS DIFFERENT

Not just “chat with a database” The system helps users determine which research questions are supportable by the available data and how the dataset can be used to address them.	Designed for multidisciplinary research It connects clinical questions to data discovery, enabling investigators, clinicians, analysts, and biostatisticians to work from a shared understanding of the dataset.
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IMPACT & ST. JUDE RELEVANCE

Investigators & clinicians Start with science and quickly understand what the dataset can support.	Analysts & biostatisticians Locate needed information, assess data availability, and understand how data elements connect.	Research programs Create a reusable pathway for navigating complex resources such as CCSS and SJLIFE.
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CURRENT STATUS

Proof-of-concept architecture demonstrated using MIMIC-IV with structured SQL retrieval and RAG over clinical text.
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Start with the scientific question. Let the system help find the right data.